

LESSON 8-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Writing Inequalities**

Maryland Standard 6.EE.B.8 • Learning target – I can write an inequality that describes a constraint in a real situation.

REMEMBER

- An **inequality** compares amounts using $<$, $>$, $\leq$, or $\geq$.
- **more than:** $>$ **less than, fewer than:** $<$
- **at least:** $\geq$ **at most, no more than:** $\leq$
- “At least” and “at most” **include** the boundary.

WORKED EXAMPLE

A food pantry needs at least 40 volunteers on Saturday. Write an inequality for v , the number of volunteers.

Step 1 “At least 40” means 40 or more, so 40 is included.

Step 2 Use $\geq$: $v \geq 40$. Test: 40 and 52 work; 38 does not.

Answer $v \geq 40$

WORD BANK

at least – that number or more ($\geq$)

at most / no more than – that number or less ($\leq$)

more than – greater, not equal ($>$)

fewer than / below – smaller, not equal ($<$)

PRACTICE*show your work • label units*

1. Write an inequality for each phrase. Use x .

Step 1 Find the key words.

Step 2 Is the number itself allowed? Yes: $\leq$ or $\geq$. No: $<$ or $>$.

a. more than 12 _____ **b.** at most 30 _____ **c.** fewer than 8 _____

d. at least 15 _____ **e.** no more than 6 _____ **f.** greater than -3 _____

g. below 0 _____ **h.** a maximum of 100 _____

2. Write an inequality for each sign. Use the variable given.

Sign	Inequality
Riders must be at least 54 inches tall. (h)	
Speed limit: 35 miles per hour. (s)	
Buy more than 10 items to get a discount. (n)	
Fewer than 120 people may be in this room. (p)	

3. **City Bus.** A city bus can carry **no more than** 48 riders. Write an inequality for r , the number of riders. Can the bus carry 48 riders? Can it carry 50 riders? Explain.

Step 1 "No more than" means that number or less.

Step 2 Test 48. Then test 50.

Inequality _____

4. **Reverse It.** Write a real-world situation for $t < 60$. Name two solutions.

Step 1 $t < 60$ means t is less than 60.

Step 2 Think of something that must stay under 60, like minutes.

Situation and solutions _____

5. **Same or Different?** Jordan says "x is at most 10" and "x is less than 10" mean the same thing. Is he right? Use the number 10 to explain.

Frame Jordan is ___ because 10 ___ a solution of $x \leq 10$, but 10 ___ a solution of $x < 10$.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Look for a sign with a limit (speed limit, height rule, room capacity). Write an inequality for it and tell someone at home what it means.

LESSON 8-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Representing Inequalities**

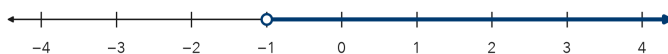
Maryland Standard 6.EE.B.8 • Learning target – I can represent an inequality with a set of solutions and describe them.

REMEMBER

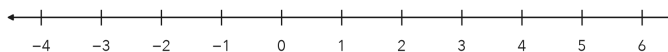
- **Open dot:** boundary **not** included ($<$ or $>$).
- **Closed dot:** boundary **is** included ($\leq$ or $\geq$).
- Arrow: **left** for $<$ and $\leq$, **right** for $>$ and $\geq$.

WORKED EXAMPLEGraph $x \leq 3$.**Step 1** $\leq$ includes 3, so draw a closed dot at 3.**Step 2** Shade left: the solutions are 3 and less.**Answer** Closed dot at 3, arrow left**WORD BANK****open circle** – the number is not a solution**arrow** – points to all the solutions**closed circle** – the number is a solution**PRACTICE***use the number lines • show your work*

1. Write the inequality shown on the number line. Use x .

Step 1 Find the number at the circle. Is it open or closed?**Step 2** Which way does the arrow point? Right means greater.**Inequality** _____

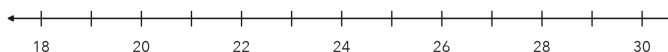
2. Graph $x \geq 2$ on the number line. Then name two solutions.

Step 1 $\geq$ includes 2, so the circle is closed.**Step 2** Draw the arrow toward the greater numbers.**Two solutions** _____

3. Each graph is described in words. Write the inequality for each graph. Use x .

a. open dot at 3, arrow right _____**b.** closed dot at -2, arrow left _____

4. **Water Bottle.** A bottle holds at most 24 ounces of water. Write an inequality for w , the ounces of water. Graph it.

Step 1 "At most" means that number or less.**Step 2** Choose open or closed. Then choose the arrow direction.**Inequality** _____

- 5. Find the Error.** Keon graphed $n > 5$ with a closed dot at 5 and an arrow to the right. Explain his mistake. Is 5 a solution?

Frame Keon should use an ___ circle because 5 ___ a solution of $n > 5$.

TEST PRACTICE

choose the best answer

- 6.** Which inequality is shown on the number line?



A $x < 4$

B $x \leq 4$

C $x > 4$

D $x \geq 4$

- 7.** Which number is a solution of $n > -3$?

A -5

B -4

C -3

D 0

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #___